

5-Why Analysis — US Flight Delays 2023

Project: Flight Delay Root Cause Analysis (DMAIC)

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Problem Statement:

20.06% of US domestic flights arrived more than 15 minutes late in 2023

(DPMO: 200,600 | Sigma Level: ~2.3)

Root Cause #1: Late Aircraft Delay (avg 28.47 min — highest of all causes)

Why 1: Why do flights arrive late?

→ The arriving aircraft was late from its previous flight segment.

Why 2: Why was the previous flight late?

→ Turnaround time at the origin airport was insufficient to recover the schedule.

Why 3: Why was turnaround time insufficient?

→ Boarding and deplaning took longer than the scheduled ground time allowed.

Why 4: Why did boarding/deplaning exceed scheduled time?

→ Airlines schedule minimum ground times to maximize aircraft utilization, leaving no buffer for operational disruptions.

Why 5: Why is there no buffer built in?

→ ROOT CAUSE: Airline scheduling practices prioritize aircraft utilization over schedule reliability, creating a cascading delay effect across the entire daily flight network — especially during peak summer months

when FLL, MCO, and MIA experience compounded weather + demand pressure.

Root Cause #2: Carrier Delay (avg 25.84 min — second highest)

Why 1: Why do airlines cause their own delays?

→ Aircraft maintenance issues discovered at gate delay pushback.

Why 2: Why are maintenance issues found at the gate?

→ Preventive maintenance schedules are insufficient to catch all faults before departure.

Why 3: Why are maintenance schedules insufficient?

→ Tight aircraft rotation schedules leave minimal time for thorough pre-flight inspections.

Why 4: Why are rotations so tight?

→ Fleet utilization targets prioritize revenue hours over maintenance windows.

Why 5: Why are maintenance windows not protected?

→ ROOT CAUSE: Airline operational models are optimized for cost efficiency, not reliability owing to reactive (not preventive) maintenance culture, particularly at high-utilization carriers like B6 and F9.

Data Evidence (from SQL analysis):

- Late Aircraft = 28.47 min avg (largest delay cause)
- Carrier = 25.84 min avg (second largest)
- JetBlue (B6) on-time rate: 69.2% (worst in industry)

- Frontier (F9): 69.24% (second worst)
- July 2023 on-time rate: 71.93% (worst month — peak demand + weather)
- FLL (Fort Lauderdale): 69.37% on time (worst airport)